

Uma Shankar

Data Engineer

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SUMMARY

Results-driven Data Engineer with 5+ years of experience designing, building, and optimizing large-scale data pipelines and analytics solutions across diverse industries. Proven expertise in ETL development, data modeling, cloud platforms (AWS, Azure, GCP), distributed systems, and big data frameworks. Skilled in Python, SQL, Apache Spark, Airflow, Kafka, Snowflake, and CI/CD pipelines, delivering scalable, secure, and business-impactful data solutions. Adept at collaborating with cross-functional teams to translate business needs into high-quality, production-grade data products.

PROFESSIONAL EXPERIENCE

Data Engineer HCL (Internship) USA	Jan 2025 – Current
- Designed scalable, high-performance ETL pipelines using PySpark, AWS Glue, and Apache Airflow, orchestrating automated ingestion, transformation, and loading of structured and unstructured data from diverse sources, reducing end-to-end processing time by 35% and enabling near-real-time analytics for business-critical reporting. - Built and maintained an enterprise-grade data lake architecture on AWS S3 with optimized partitioning strategies, lifecycle policies, and compression formats (Parquet, ORC), achieving a 20% reduction in storage costs while ensuring faster query execution. - Developed dimensional data models (Star/Snowflake schemas) and materialized views in Snowflake to support BI and analytics teams, improving query performance by 40% and enabling self-service reporting capabilities. - Integrated real-time streaming data pipelines from Apache Kafka and AWS Kinesis into Amazon Redshift, implementing transformation logic for event-based processing, reducing operational reporting delays and enabling instant KPI tracking. - Created and deployed automated data quality frameworks in Python with AWS Lambda triggers to perform schema validation, null checks, duplicate detection, and business rule validations, ensuring 99% accuracy across enterprise reports. - Implemented CI/CD deployment pipelines using Jenkins and Git for ETL jobs, Airflow DAGs, and Snowflake SQL scripts, reducing deployment times from hours to under 30 minutes and ensuring zero downtime releases. - Collaborated with data science teams to design reusable feature engineering pipelines for machine learning models, incorporating advanced aggregations and time-series transformations, resulting in a 15% improvement in model prediction accuracy. - Applied data governance and compliance measures including GDPR, HIPAA, and encryption-at-rest/in-transit (KMS, SSL), implementing RBAC in Snowflake to protect sensitive data and pass all security audits. - Partnered with stakeholders to define KPIs and deliver interactive BI dashboards in Power BI and Tableau, directly connected to Snowflake datasets, reducing ad-hoc report requests by 50% and improving decision-making speed. - Operated in an Agile/Scrum environment, leading sprint planning, backlog grooming, daily stand-ups, and retrospectives to ensure timely delivery of high-priority data engineering projects.	

Data Engineer TCS India	July 2018 – July 2023
- Designed and deployed large-scale data ingestion frameworks using Apache Spark, Hadoop, and Hive to process 10TB+ of structured, semi-structured, and unstructured datasets daily, improving data pipeline efficiency and scalability. - Led the migration of an on-premises Oracle data warehouse to Azure Synapse Analytics, building automated ETL processes in Azure Data Factory and achieving a 25% reduction in infrastructure costs while improving query performance. - Built incremental and real-time ETL workflows in Azure Data Factory to process streaming data feeds with a latency of under 2 minutes, enabling near-real-time operational dashboards. - Developed Python-based metadata management and lineage tracking automation scripts, integrating with Azure Purview for improved data governance, regulatory compliance, and faster issue resolution. - Created OLAP cubes and optimized dimensional data models to support high-speed BI queries in tools like Power BI and Tableau, reducing report generation time by 30% and improving user adoption. - Integrated external data sources through third-party APIs, performing enrichment and harmonization with existing enterprise datasets to enhance the accuracy and completeness of advanced analytics initiatives. - Performed Hadoop cluster optimization and Spark job tuning, improving data processing performance by 20% through executor tuning, shuffle optimization, and memory configuration enhancements. - Developed role-based access controls (RBAC) and encryption policies for sensitive datasets in Hive and Azure Synapse, ensuring compliance with corporate data security standards and industry regulations. - Collaborated with QA and UAT teams to validate ETL output through structured test cases, data reconciliation scripts, and thorough defect resolution processes, reducing post-deployment data anomalies by over 90%. - Mentored junior data engineers on SQL query optimization, PySpark transformations, and data modeling best practices, improving overall team productivity and reducing onboarding time for new hires.	

TECHNICAL SKILLS

- Data Analysis & Visualization:** Python, SQL, Java, Scala, PySpark, C Python (NumPy, Pandas), Shell Scripting, R
- Data Engineering:** Python, SQL, MongoDB, Grok, APIs, Regex, JSON, YAML, Azure Data Factory (ADF), Databricks, AWS, Azure DevOps, SFTP, SDLC, Docker, Kubernetes, CloudWatch
- Big Data & Machine Learning:** Apache Spark, Hadoop, Hive, Scikit-Learn, TensorFlow, Keras, Hadoop, HDFS, Scala, SAP HANA
- Cloud Platforms:** AWS, AWS GLUE, AWS Redshift, AWS Lambda, Google Cloud Azure Databricks, Data Lake (ADLS), Cosmos DB, AWS S3, EMR, DynamoDB
- Data Warehousing & ETL:** ETL Development, Data Modeling, Data Lakes, Amazon Redshift, Google Big Query, Azure Data Factory, Azure Databricks, Azure Synapse, Snowflake
- Data Visualization & BI Tools:** Tableau, Power BI, Microsoft Office Suite, Power Pivot, Power View, Power Query
- Databases:** MySQL, PostgreSQL, NoSQL (MongoDB, Cassandra), Oracle, Teradata, SQL Server, PL/SQL, BigQuery
- Version Control & CI/CD:** Git, Jenkins, Docker
- Data Management:** Data Migration, Data Cleansing, Data Integration, SQL, ETL Processes
- Tools & Technologies:** Microsoft Visio, Tableau, Power BI, MuleSoft, Excel (Advanced), SAP BI, SQL, Tableau, Power BI
- Project Management:** Agile methodologies (Scrum, Kanban), Jira, Confluence, ServiceNow, Scrum, sprint planning, backlog management

EDUCATION

Masters University of the Pacific, USA	Aug 2023 – May 2025
Bachelors Lovely Professional University, India	Aug 2016 - May 2020